



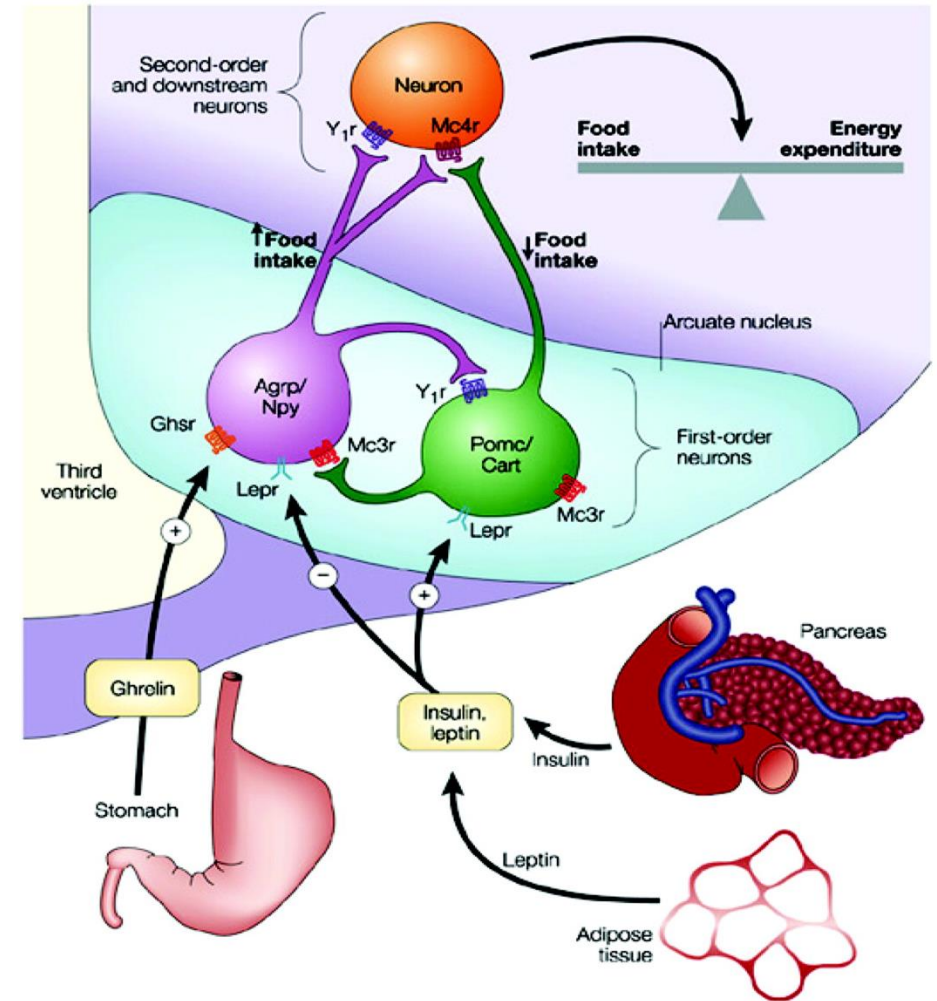
Male offspring of undernourished dams are protected from high fat diet-induced weight gain and display sex-specific metabolic changes

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Hendrix College

Lab of Drs. Gwen Childs and Angela Odle
Assisted by Dr. Tiffany Miles

Leptin: Overview

- Leptin is a hormone secreted by adipose tissues and acts as an appetite suppressor in the hypothalamus, regulating food intake
- Circulating leptin levels are proportional to adipose storage
- Increased circulation of leptin is associated with obese individuals, who have higher adiposity
- Leptin receptors are also found in the pituitary



Foster-Schubert, K. E., & Cummings, D. E. (2006). Emerging therapeutic strategies for obesity. *Endocrine reviews*, 27(7), 779-793.

Musashi In The Anterior Pituitary Is Regulated By Leptin Signaling

- In mice, where leptin receptor (LEPR^{null}) is specifically ablated in growth hormone producing cells, Msi1 mRNA is increased
- Musashi
 - RNA binding protein
 - Two isoforms: MSI1 and MSI2
 - Translational regulator: prevents or promotes translation of RNA to protein
 - Translation of Pou1f1, which is necessary for pituitary maturation, is inhibited by MSI

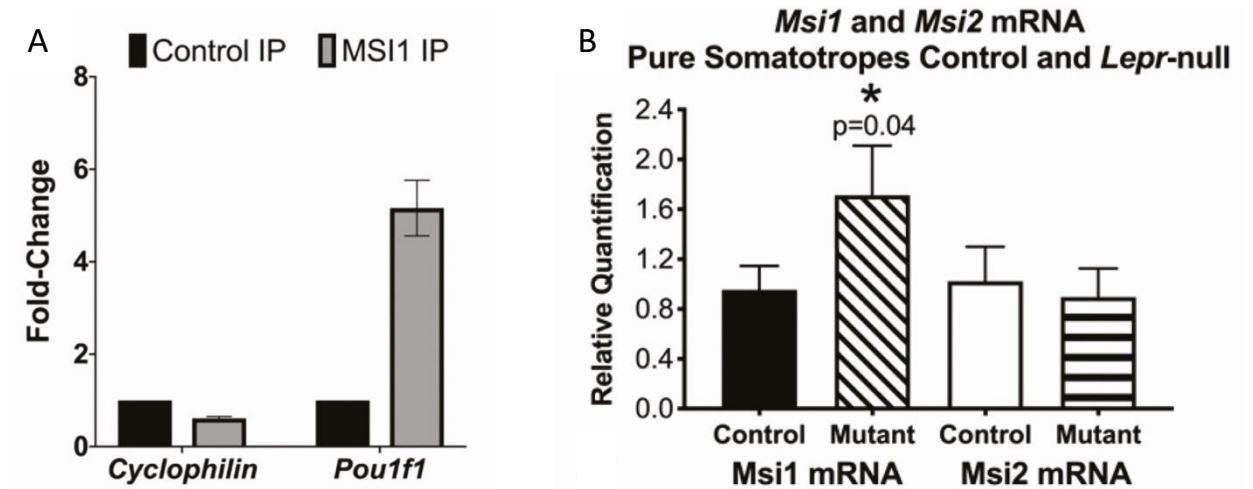


Figure 2: A) *Msi1* transcript increased with loss of leptin signaling. B) Pou1f1 is a target of MSI shown by immunoprecipitation

Source

Allensworth-James, M., Banik, J., Odle, A., Hardy, L., Lagasse, A., Moreira, A.R.S., Bird, J., Thomas, C.L., Avaritt, N., Kharas, M.G. and Lengner, C.J., 2021. Control of the anterior pituitary cell lineage regulator POU1F1 by the stem cell determinant Musashi. *Endocrinology*, 162(3), p.bqaa245.

Neonatal Leptin Surge

- A surge in leptin is seen within the neonatal development of both rodents and humans
 - 3rd trimester in humans
 - Postnatally in rodents
- Occurs independent of adiposity and prompts development of anorexigenic (appetite suppressant) neurons
- Studies suggest that maternal undernutrition alters the levels of the leptin surge, making the surge happen prematurely.
 - Premature leptin surge led to metabolic dysfunction and weight gain in adult progeny.

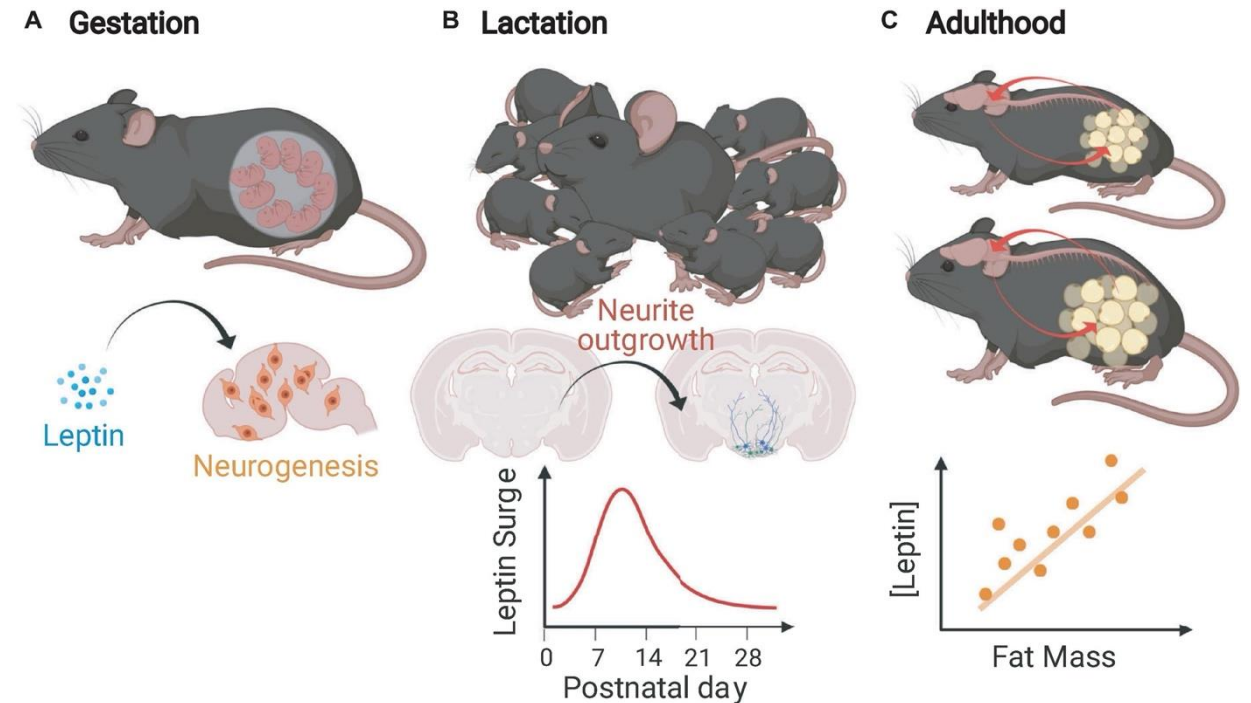


Figure 4: Graphs of Neonatal Leptin Surge
Source: <https://www.frontiersin.org/articles/10.3389/fphys.2021.789519/full>

What we want to know...

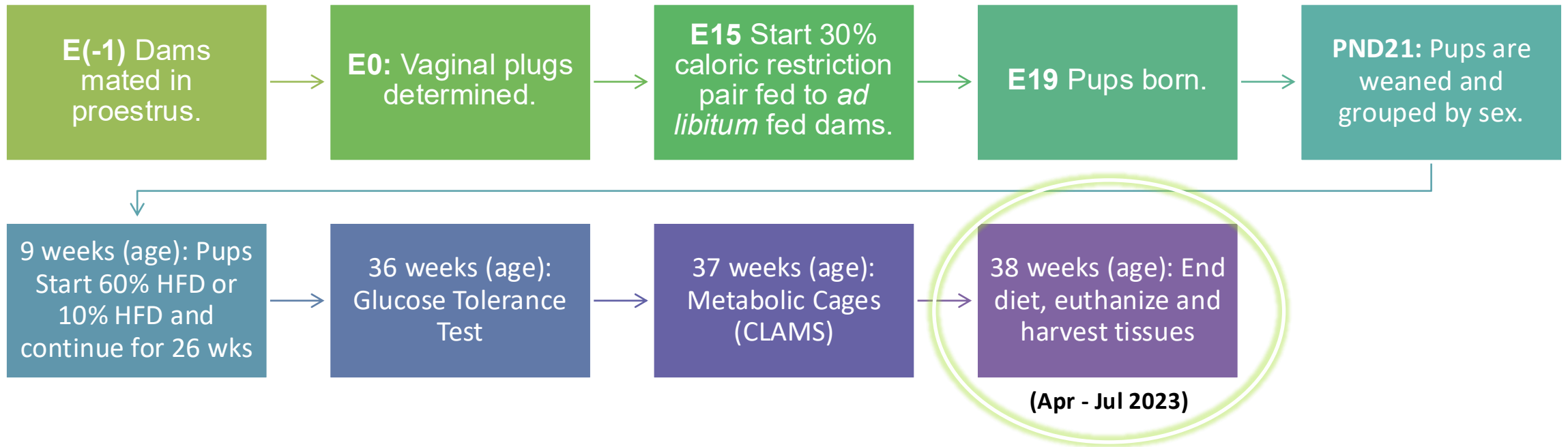


We will be focusing on altering **leptin** itself and understanding its affects on pituitary development and metabolic function



Hypothesis: A 30% maternal undernutrition in FVB mice will blunt the neonatal leptin surge and will alter MSI target expression in neonates that will persist in the adult offspring.

(Aug - Nov 2022)



- Mice strain: FVB.129P2-Fmr1tm1Cgr/J, "Sighted FVB"
- Part 1: Maternal mice were fed a 30% calorie restricted diet (CR30) to cause a blunted leptin surge in their progeny.
- **Part 2: These progeny were then fed either a control diet or a high fat diet. Weekly weight gain was assessed along with serum leptin and metabolic functions such as glucose tolerance testing, and metabolic cages in adult progeny.**

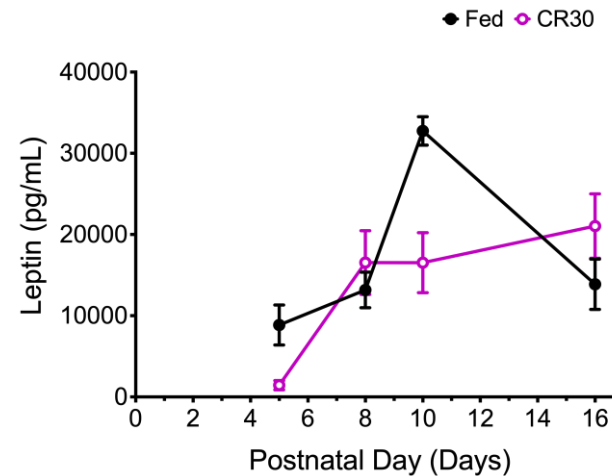
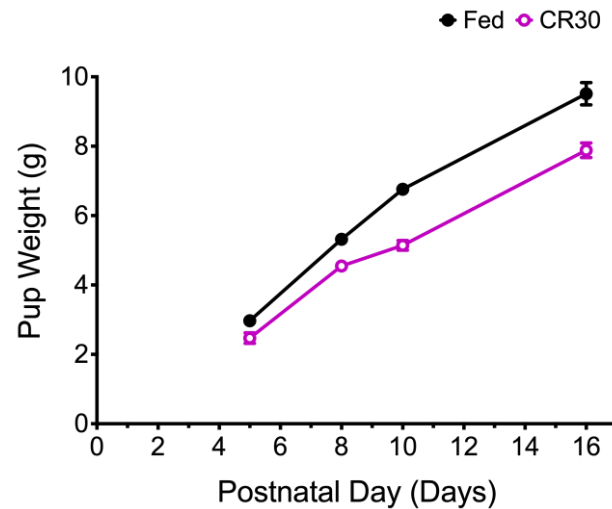
CR30 Dams = 5

- Males CTRL = 6
- Males HFD = 6
- Female CTRL = 9
- Female HFD = 9

Fed Dams = 5

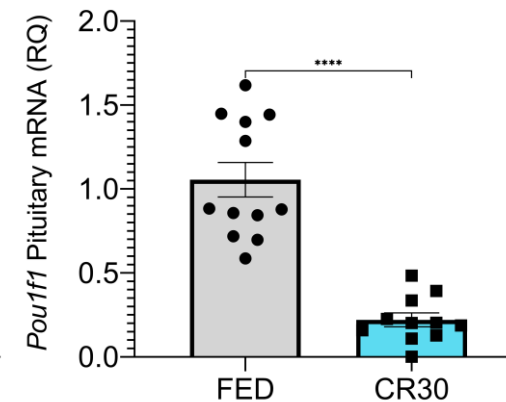
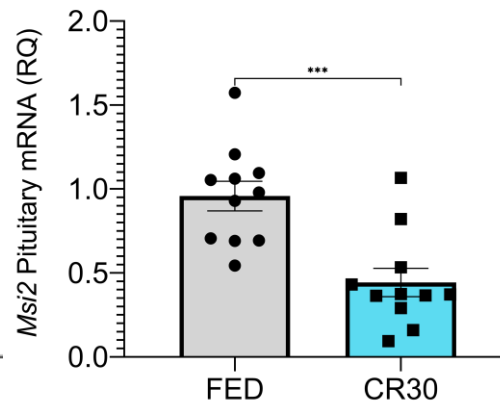
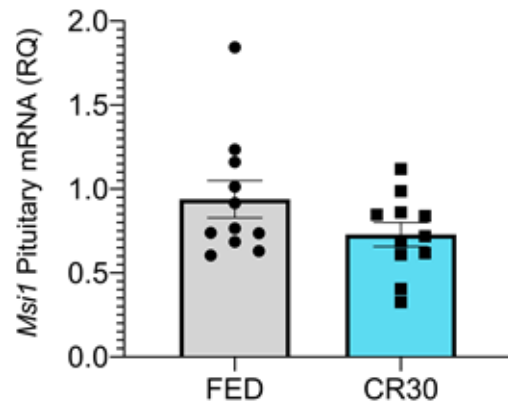
- Males CTRL = 8
- Males HFD = 9
- Female CTRL = 11
- Female HFD = 11

Part 1: Moderate (30%) maternal undernutrition in FVB mice blunts neonatal leptin surge and alters *Msi* and *Pou1f1* mRNA

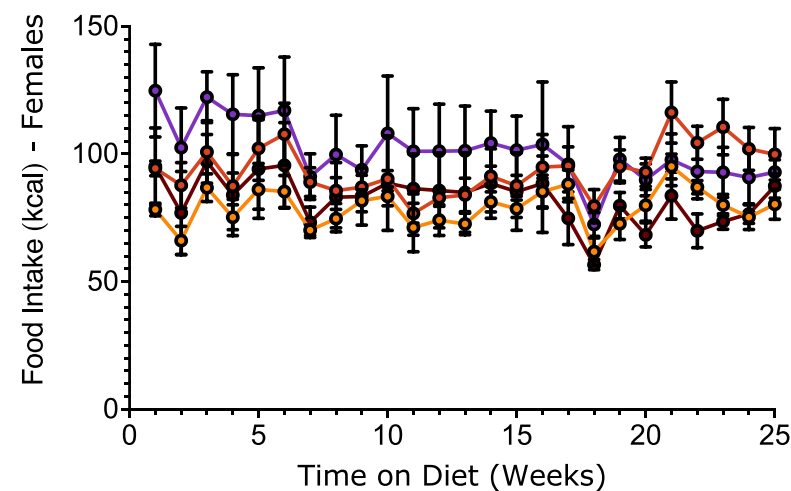
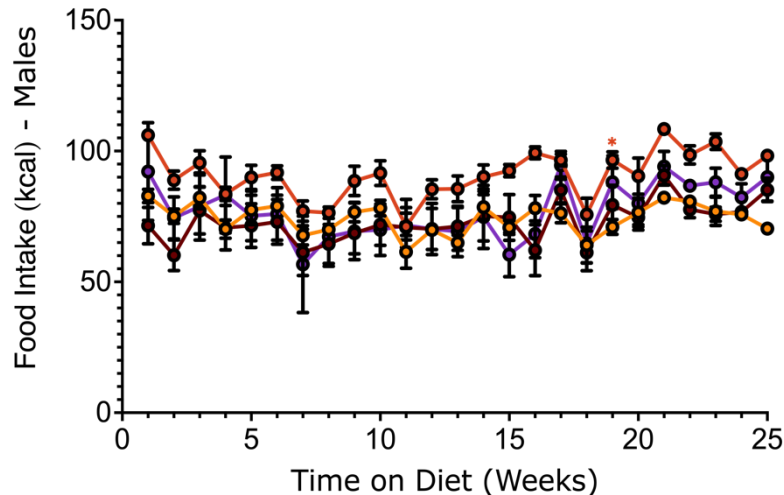
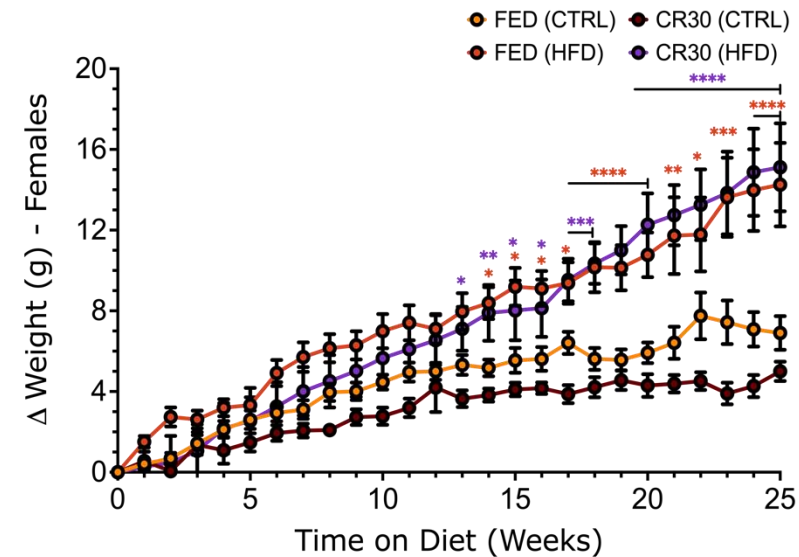
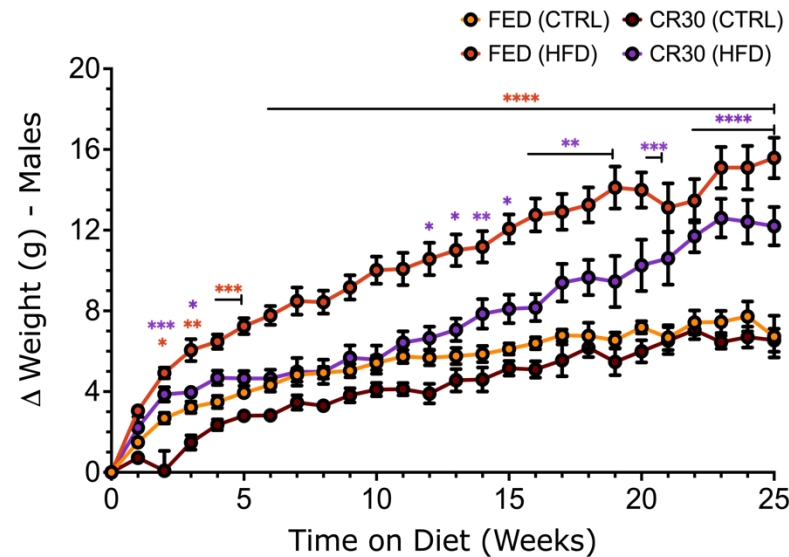


	Fed	CR30
PND 5	10	7
PND 8	22	13
PND 10	15	12
PND 16	18	13
TOTAL	65	45

PND 10



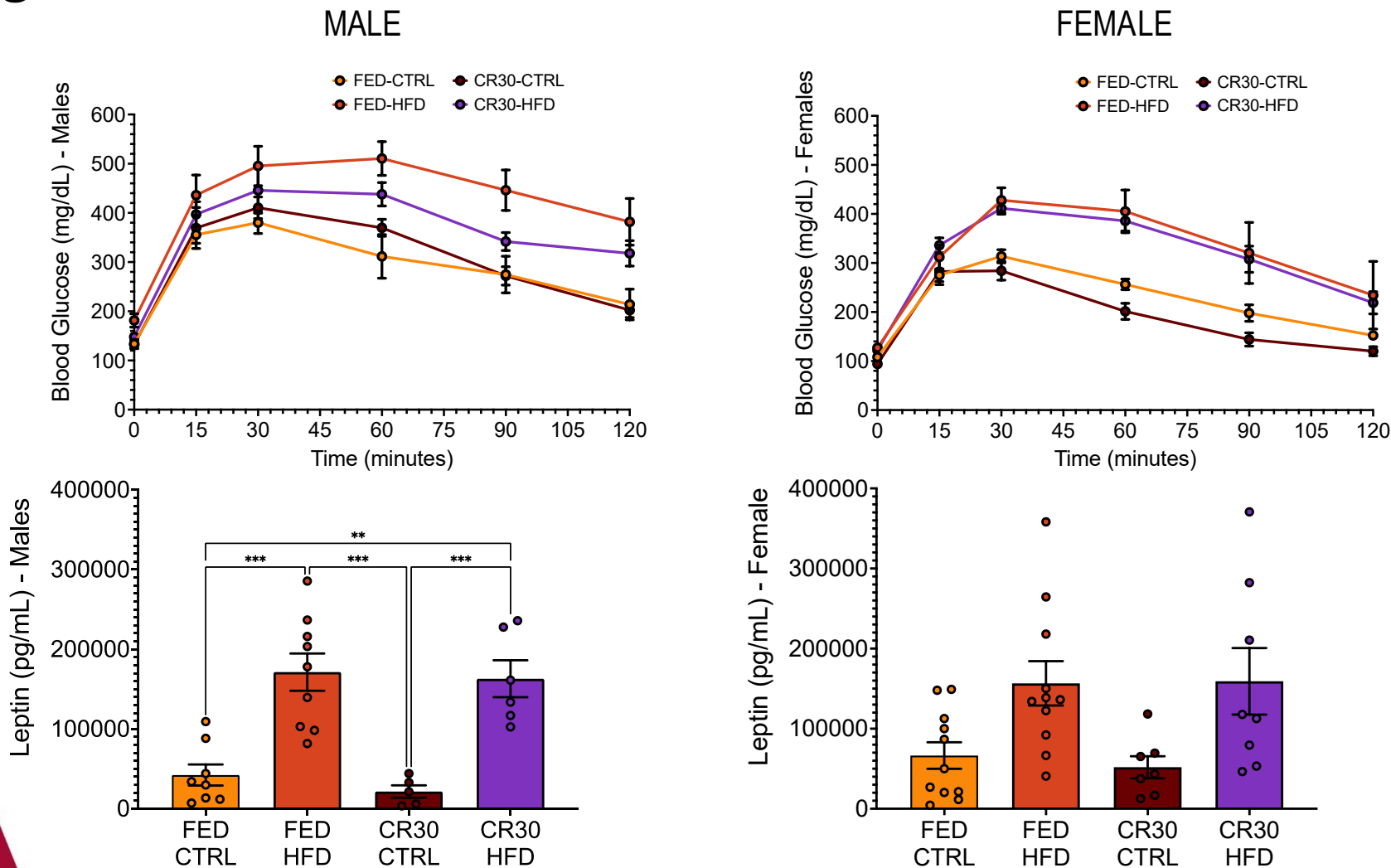
Part 2: CR30 HFD males weekly weight gain was blunted, but not the CR30 HFD females



Glucose Tolerance Testing

- GTT is a metabolic tolerance test used to determine the body's efficiency of using and storing glucose
 - Used to diagnose conditions such as insulin resistance and diabetes mellitus
 - Obesity is a common condition for impaired glucose clearing within the body
- GTT Procedure in mice:
 - Mice are fasted for 6 hrs. prior to testing
 - A baseline glucose measurement (T0) is taken using a handheld glucometer. A small drop of blood is extracted from the tails of the mice
 - Injection of glucose is then administered: $\text{Volume} = 20\% \text{ glucose stock solution} \times \text{body weight (g)}$
 - Blood glucose is then evaluated every 15-, 30-, 60-, 90-, and 120-minutes post injection.

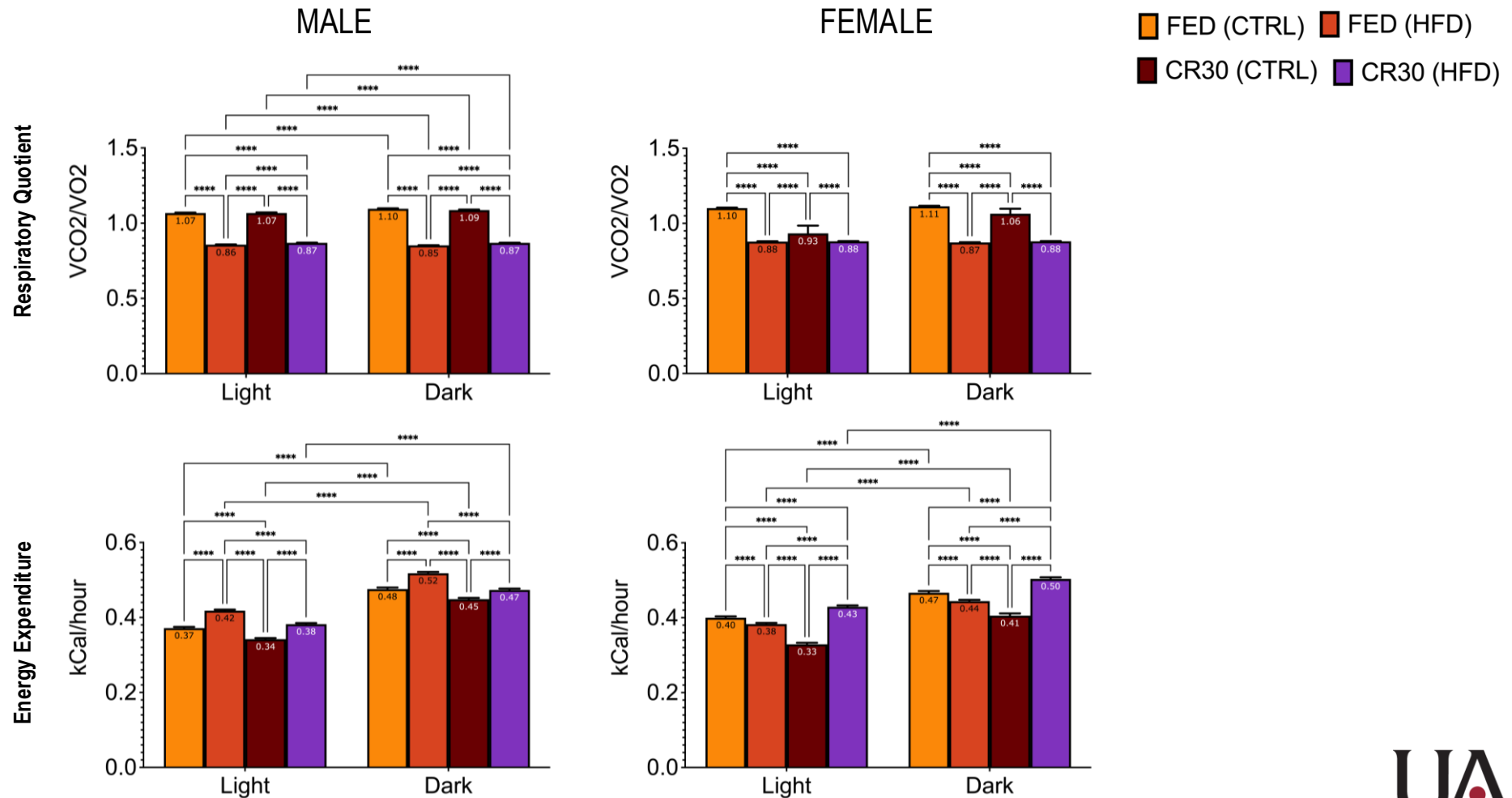
Part 2: CR30 HFD males had a reduced spike in glucose levels and clearance compared to CR30 HFD females. Serum leptin trended with weight gain.



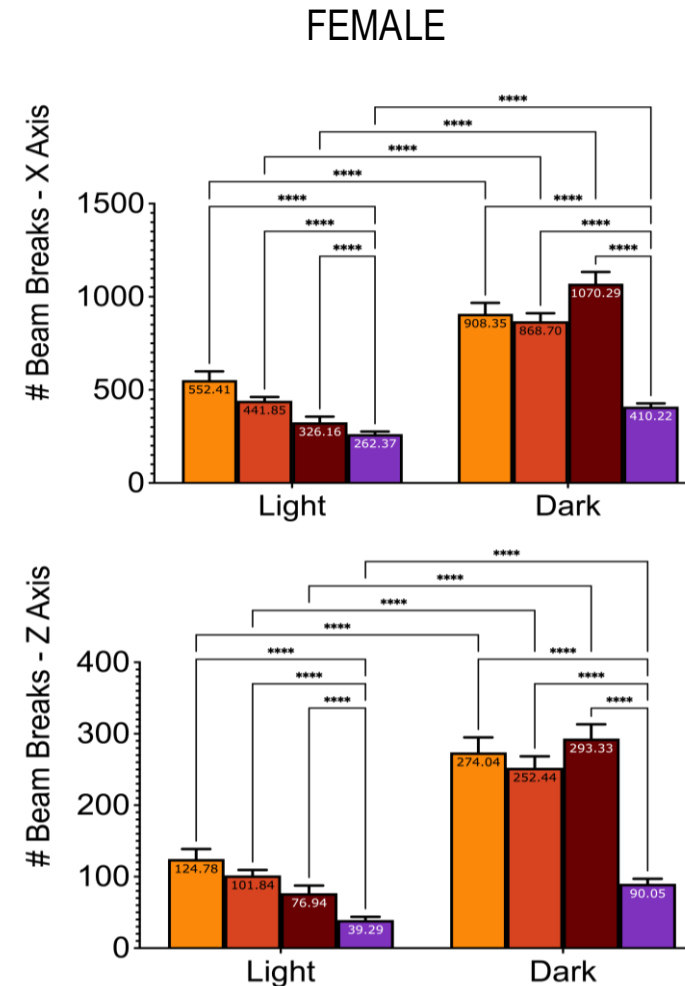
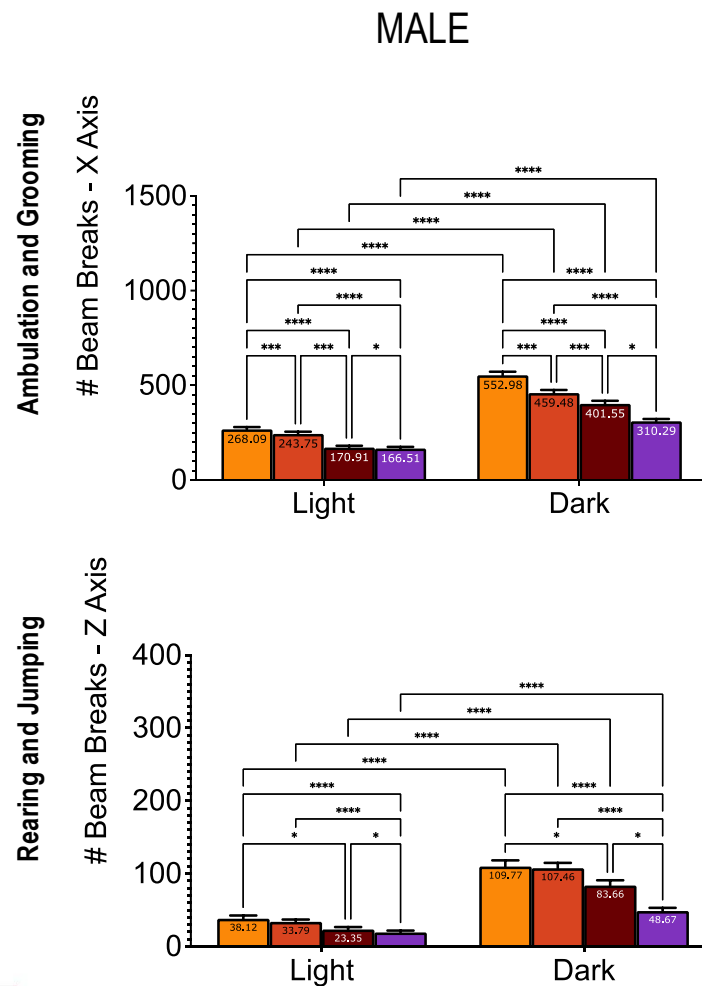
CLAMS: Comprehensive Lab Animal Monitoring System

- Measures/Calculates
 - Respiratory Exchange Ratio (RER)
 - Indicates preferential energy source
 - Values close to 1.0 = carbohydrates are main energy source
 - Values closer to 0.7 = fats are main energy source
 - Energy Expenditure: body heat
 - Physical Activity: movement on X- and Z- axis
- Assessed at thermoneutral temperature

Part 2: RER is reduced (~ 0.87) in all HFD group, while energy expenditure was increased in all HFD groups except for Fed-HFD females.



Part 2: Ambulation and Grooming (X-axis) and Rearing and Jumping (Z-axis) movement was reduced among CR30-HFD groups



■ FED (CTRL)
 ■ FED (HFD)
 ■ CR30 (CTRL)
 ■ CR30 (HFD)

- Among Fed-HFD group, males had reduced movement along X-axis, and not females
- X- and Z-axis movement was increased in CR30-CTRL females during the dark phase

Conclusions

	Expected	Experimental Conclusions
Weight	HFD group from maternal undernourished dams gain more weight on the HFD compared to Fed-HFD groups. (Thrifty Phenotype Hypothesis)	CR30 adult males did not gain as much weight on HFD compared to Fed-HFD males, whereas females were not protected from HFD weight gain.
GTT	HFD fed mice produce higher glucose spikes in response to injection and take longer to metabolize and clear glucose from the blood.	CR30 males appear to have a reduced glucose curve compared to Fed-HFD males, suggesting increased insulin sensitivity as a result of slower weight gain compared to Fed-HFD
CLAMS	Trends in RER and EE match previous studies in obese FVB mice.	CR30-CTRL have reduced mobility and further reduced mobility with HFD. This is an unexpected result since reduced mobility would promote weight gain (as seen in females) but does not account for slower weight gain in males.

These experiments have further characterized the phenotype.
Now, mechanistic data is needed!

Next Steps

Proteo-genomics of MSI targets in neonates and adults to determine the effect of this maternal undernutrition model (blunted neonatal leptin surge) on MSI action in the pituitary.



Thank You & Questions

University of Arkansas for Medical Sciences

- Dr. Gwen Childs
- Dr. Angela Odle
- Dr. Tiffany Miles
- Alex Lagasse

Summer Undergraduate Research Program (SURP) to Increase Diversity in Research

- Dr. Latrina Prince

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- 2017-2023: R01DK113776 (GC, MM, AM)
- 2022-2025: 5 R01 DK127723 (GC, MM, AM)
- NCATS TRI: UL1 TR003107 and TL1 TR003109

Adjusted Calories Diet (60/Fat)

Formula	g/Kg
Casein	265.0
L-Cystine	4.0
Maltodextrin	160.0
Sucrose	90.0
Lard	310.0
Soybean Oil	30.0
Cellulose	65.5
Mineral Mix, AIN-93G-MX (94046)	48.0
Calcium Phosphate, dibasic	3.4
Vitamin Mix, AIN-93-VX (94047)	21.0
Choline Bitartrate	3.0
Blue Food Color	0.1

Footnote

Approx. 60% of total calories come from fat. Designed with similarities to Research Diets, Inc. formula D12492. For the series TD 06414-TD 06416. Approximate fatty acid profile (% of total fat): 36% saturated, 41% monounsaturated, 23% polyunsaturated.

Selected Nutrient Information¹

	% by weight	% kcal from
Protein	23.5	18.3
Carbohydrate	27.3	21.4
Fat	34.3	60.3

Kcal/g 5.1

¹Values are calculated from ingredient analysis or manufacturer data

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Key Features

- + Purified Diet
- + Diet Induced Obesity
- + High Fat

Key Planning Information

- + Products are made fresh to order
- + Store product at 4°C or lower
- + Use within 6 months (applicable to most diets)
- + Box labeled with product name, manufacturing date, and lot number
- + Replace diet at minimum once per week
More frequent replacement may be advised
- + Lead time:
 - 2 weeks non-irradiated
 - 4 weeks irradiated

Product Specific Information

- + 1/2" Pellet or Powder (free flowing)
- + Minimum order 3 Kg
- + Irradiation available upon request

Options (fees will apply)

- + Rush order (pending availability)
- + Irradiation (see Product Specific Information)
- + Vacuum packaging (1 and 2 Kg)

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02/18/18

Purified Control Diet (10% Fat Kcal)

Formula	g/Kg
Casein	210.0
L-Cystine	3.0
Corn Starch	465.0
Maltodextrin	100.0
Sucrose	90.0
Lard	20.0
Soybean Oil	20.0
Cellulose	37.15
Mineral Mix, AIN-93G-MX (94046)	35.0
Calcium Phosphate, dibasic	2.0
Vitamin Mix, AIN-93-VX (94047)	15.0
Choline Bitartrate	2.75
Yellow Food Color	0.1

Footnote

Approx. 10% of total calories come from fat. A modification of TD.06416 to reduce sucrose to the level found in TD.06414 (9%) and replace with starch. Color coded yellow.

Selected Nutrient Information¹

	% by weight	% kcal from
Protein	18.6	20.5
Carbohydrate	62.6	69.1
Fat	4.2	10.5

Kcal/g 3.6

¹Values are calculated from ingredient analysis or manufacturer data

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